Subject: Project Proposal
Project Title: WanderBoard: A Collaborative, Role-Based Travel Discovery and Planning Platform
Group member's Name and Enrolment Number: Dhruv [24115057] & Keshav [24114048] & Naivadhya [24115106] & Garima [24113043] % Dakshata [24114029] & Bhoomika [24114043]

WanderBoard

The Target Customers

Our target customer represents a group of four enthusiastic friends planning a vacation, illustrating the exact chaotic dynamics our system aims to solve. Each friend brings a specific need that shapes the software's functionality:

- **A (The Lead Planner):** She takes on the stressful responsibility of evaluating the feasibility of proposed locations and finalizing the actual schedule. She represents the need for **Role-Based Access Control**, requiring a system where everyone can share ideas, but only she, as the "Trip Owner," has the authority to drag and drop items into the official timeline. To make these final decisions, she actively relies on the **Local News Feed** and **Environmental Data Dashboard** to monitor weather alerts and regional updates, ensuring the itinerary is both safe and practical before locking it in.
- **B (The Photographer):** He is bringing his DSLR and is highly focused on capturing local landscapes. He highlights the need for the **Environmental Data Dashboard** and **Smart Contingency Flagging**. He needs the system to track exact sunrise and sunset times to catch the "golden hour" and automatically warn the group if an outdoor shoot conflicts with forecasted rain.
- **C (The Experience Seeker):** He wants to explore the city's culture beyond the typical tourist traps, specifically looking for highly-rated street food markets and live music gigs happening during their stay. He illustrates the necessity of the **Smart Discovery Engine** and **Local Events Feed**, allowing him to actively search for these dynamic, time-sensitive activities and drop them into the group's shared "Sandbox" for voting.
- **D (The Pragmatist):** She manages the group's overall budget and logistics. She highlights the need for **Hotel Price Aggregation** to compare accommodation costs directly within the shared workspace, as well as an **Integrated Expense Tracker**. Since group trips often involve messy shared costs—from splitting cabs to paying for group dinners—Aisha needs the platform to seamlessly log who paid for what, split bills dynamically, and automatically calculate simplified settlement balances to avoid awkward post-trip math.

Through this narrative, the customer group will help refine the specifications, provide feedback over the course of the project, and conduct acceptance testing to ensure the final product resolves the friction of collaborative trip planning.

The Problem and Its Importance

Group trip planning is traditionally chaotic, forcing travelers to juggle dozens of browser tabs, ranging from review sites and maps to weather forecasts and group chats. While everyone in a travel group wants to discover exciting local activities, expecting one designated planner to research, aggregate, and organize all this unstructured data leads to burnout. Furthermore, without a central "source of truth," groups often miss out on hidden local gems or optimal event timings. This problem is important to solve because a fragmented, stressful planning phase often ruins the enthusiasm for the trip itself. Groups need a unified system that actively helps them discover local activities, democratizes idea-sharing, and centralizes final decision-making with context-aware travel data.

What the System Will Do

We propose a structured collaborative travel planning and discovery system designed to streamline the group vacation experience. Specifically, WanderBoard will:

- **Serve as an active discovery engine**, allowing users to explore local landmarks, dining, and upcoming events tailored to their destination and dates.
- **Host collaborative workspaces** where users can initiate a new trip and seamlessly invite friends.
- **Maintain a shared "bucket list"** for all group members to save discoveries, pitch ideas, and upvote their favorite activities.
- **Aggregate critical destination data**—such as localized weather forecasts, precise sunrise/sunset times, local news, and comparative hotel prices—directly within the shared dashboard.
- **Manage shared finances** through an integrated expense tracking module that logs costs, applies custom split ratios, and calculates simplified settlement balances to resolve group debts.
- **Provide smart contingency flagging** that automatically detects weather conflicts (e.g., an outdoor hike scheduled during rain) and suggests indoor alternatives.
- **Enforce role-based access control**, empowering the "Trip Owner" with the exclusive ability to finalize the schedule by moving voted items from the bucket list into the official daily timeline.
- *(Optional Feature)* **Generate baseline itineraries** using an AI-assisted planning module that the Trip Owner can review and customize.
- *(Optional Feature)* **Curate explorable locations** by actively extracting relevant data from popular travel blogs across the web.

Justification of Scope

This proposal is neither too easy nor too ambitious for a group of programmers to undertake in one semester. It is not too easy because it requires designing distinct user permission tiers, managing multi-user data interactions within a shared workspace, integrating multiple external data sets (environmental and pricing) into a unified dashboard, and implementing a debt-simplification algorithm for the expense tracker. However, it is not too ambitious because it focuses strictly on user-driven scheduling, organization, and internal settlement math, deliberately avoiding the overwhelming complexity of building automated geographic transit routing, real-time ride-hailing integrations, or live payment gateways.